👩‍🏫 Staff Guide · Small Engines Overhaul · Unit 5 Carburetion & Fuel (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Weeks 4-5: Carburetor Handout + Carburetor Article, Carburetor Video Handout, Carburetor Quiz, Venturi Lab, iCADEMY Bronze 8 Carburetion, iCADEMY fuel/oil topics (Stihl oil through ethanol in today's fuel)")	**Level 2 piece**
Carburetor Handout + Article (scanned)	Deck + notes A-C; L3 reading replaces the article
Carburetor Video Handout (2 YouTube videos)	Video guide
Carburetor Quiz (paper, no key in export)	Quiz + key (team-written key)
Venturi Lab (title only)	Venturi & Carburetor lab packet (team-written)
iCADEMY Bronze 8 + fuel/oil/ethanol	Our own slides 'Fuel and Oil Care' (no STIHL content)
Class slides	Deck: 15 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 9 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Clear carburetor model or cutaway if available
- Straw-and-paper demo of the venturi

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Carburetor Quiz: in class the quiz is open-notes; let students use their guided notes
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say "check with the class teacher".

🚧 Under construction (check with the class teacher)

- **Venturi Lab**: Title only in Canvas; our lab is team-written.
- **Carburetor Article**: Scanned image, source unknown; our reading replaces it.
- **Carburetor Quiz key**: The paper quiz has no key in the export; our key is team-written.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩🏾‍🏫 Staff Guide · Venturi & Carburetor Lab (Level 2)

Small Engines Overhaul · Unit 5 · goes with the class “Venturi Lab + Carburetor Handout” (lab points). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Venturi Lab + Carburetor Handout”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (lab points)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: The class Venturi Lab is title-only in Canvas; this lab is team-written.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

⛽ Carburetor Notes

Small Engines Overhaul · Unit 5 Carburetion & Fuel · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Purpose

1. The carburetor mixes _____ and fuel.
2. The _____ controls engine speed.
3. The _____ helps a cold engine start.

Word bank: **air** **choke** **throttle**

B. Venturi

4. Air speeds up in the narrow _____.
5. Fast air has lower _____, so fuel is pushed up the jet.

Word bank: **pressure** **venturi**

C. Parts and fuel

6. Match the part to its job:

Float	pumps fuel at any angle
Jet	keeps stored fuel fresh
Diaphragm	keeps fuel level
Stabilizer	meters fuel

Level 2 version of the gen ed “Carburetor Handout + Carburetor Quiz.”

📖 Carburetion & Fuel: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Carburetor

The part that mixes air and fuel in the right amounts for the engine.



Venturi

A narrow part of the carburetor throat where air speeds up and pressure drops, pulling fuel in.



Choke

A plate that limits air for a richer mix when starting a cold engine.



Throttle

A plate that controls how much air-fuel mix enters, changing engine speed.



Float

A part that floats on fuel in the bowl and shuts off fuel when the bowl is full.



Ethanol

Alcohol mixed into gasoline (E10). It attracts water and can gum up small engines.



Fuel stabilizer

An additive that keeps stored gas from going bad.



Air-fuel ratio

The mix of air to fuel, about 14.7 parts air to 1 part fuel by weight.

⏳ Venturi & Carburetor Lab

Small Engines Overhaul · Unit 5 · Level 2



Name: _____

Modified from the class “Venturi Lab + Carburetor Handout”.

🧺 Supplies

- ☐ 🥤 **Straws** 2 per student
- ☐ 💧 **Cup of water**
- ☐ 📄 **Paper strip**
- ☐ 💧 **Practice carburetor** off the engine
- ☐ 🧼 **Carb cleaner** staff spray, ventilated
- ☐ 🥣 **Parts tray**

⏳ Steps: check each box when you finish

1		Blow across a paper strip; record what it does.	☐
2		Straw in water: blow across the top with a second straw; record.	☐
3		Explain: fast air = low pressure.	☐
4		On the practice carburetor, find the venturi, jet, choke and throttle.	☐
5		With adult supervision, remove the bowl; find the float and needle.	☐
6		Staff spray carb cleaner in a ventilated area; you watch.	☐
7		Reassemble; check the choke and throttle move freely.	☐

🚧 **Under construction:** The class Venturi Lab is title-only in Canvas; this lab is team-written. *Check with the class teacher.*

⏳ Venturi & Carburetor Lab · Prep Sheet

Small Engines Overhaul · Unit 5 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🔧 **Mechanic**	Does the main engine job with the tools	
🧰 **Tool helper**	Gets tools from the board and returns them to staff	
📋 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Wipes the bench, sweeps, checks tools are back	

🧰 Tools and supplies: check when you have it

- ☐ 🥤 **Straws** 2 per student
- ☐ 💧 **Cup of water**
- ☐ 📄 **Paper strip**
- ☐ 💧 **Practice carburetor** off the engine
- ☐ 🧼 **Carb cleaner** staff spray, ventilated
- ☐ 🥣 **Parts tray**

⚠️ Safety and manners focus

🥽 **Safety glasses** on the whole time.

👖 **Coveralls** and closed-toe shoes; tie back long hair.

🔧 Use tools **with adult supervision**. Right tool for the job.

🧼 Carb cleaner is a **chemical**: staff spray it, ventilated, away from eyes.

🧼 **Clean up** and wash hands when you finish.

✅ Before you start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

⏳ Venturi & Carburetor Lab · Reflection

Small Engines Overhaul · Unit 5 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🥽 Wore safety glasses and used tools safely

☐ 🔧 Finished the Venturi & Carburetor Lab steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about engines?

1. _____
2. _____

Ideas: **The engine needs…** **I saw…** **Next time…**

✅ Carburetion & Fuel Quiz

Small Engines Overhaul · Unit 5 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 💧 The main job of a carburetor is…

A. Making spark | B. Mixing air and fuel | C. Cooling oil

2. ⏳ A venturi is…

A. A fuel tank | B. A spark part | C. A narrow spot that speeds up air

3. 📉 Fast-moving air has…

A. Lower pressure | B. Higher pressure | C. No pressure

4. 🫧 Which carburetor needs a fuel pump / diaphragm?

A. Float type | B. Gravity type | C. Diaphragm type

5. 🚪 The choke is used for…

A. Cold starting | B. Stopping | C. Cooling

6. 🎚️ The throttle controls…

A. Oil level | B. Speed | C. Spark gap

7. 🛟 The float…

A. Makes noise | B. Shuts off fuel when the bowl is full | C. Holds oil

8. 🌽 Ethanol in fuel can…

A. Attract water and gum carburetors | B. Make engines last forever | C. Replace oil

9. 🛢️ A 2-stroke needs…

A. No oil | B. Diesel | C. Oil mixed in the fuel

10. 🌬️ Blowing across a straw in water lifts the water because…

A. Fast air lowers pressure | B. Water is light | C. Magic

🎬 Carburetor Videos

Small Engines Overhaul · Unit 5 · Level 2 video guide (modified). Two YouTube videos linked in Canvas (watch with captions).

Name: _____

Before you watch

What do you think a carburetor does? _____

While you watch: circle the best answer

1. What is the first step to remove the carburetor?

A. Start the engine | B. Shut off/drain fuel and remove the air filter | C. Remove the piston

2. What cleans the jets?

A. Water | B. Sand | C. Carburetor cleaner and air

3. What part sets the fuel level?

A. The float | B. The choke | C. The muffler

4. Why take photos while taking it apart?

A. For fun | B. No reason | C. To put the linkages back right

After you watch: finish the sentence

5. After watching, finish the sentence:

A dirty carburetor makes the engine _____

Goes with the class “Carburetor Video Handout” (class points pts). Same points when complete.

⛽ Carburetor Notes · ANSWER KEY

Small Engines Overhaul · Unit 5 Carburetion & Fuel · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Purpose

1. The carburetor mixes **air** and fuel.
2. The **throttle** controls engine speed.
3. The **choke** helps a cold engine start.

Word bank: **air** · **choke** · **throttle**

B. Venturi

4. Air speeds up in the narrow **venturi**.
5. Fast air has lower **pressure**, so fuel is pushed up the jet.

Word bank: **pressure** · **venturi**

C. Parts and fuel

6. Match the part to its job:

(Key: Float → keeps fuel level · Jet → meters fuel · Diaphragm → pumps fuel at any angle · Stabilizer → keeps stored fuel fresh)

Float	pumps fuel at any angle
Jet	keeps stored fuel fresh
Diaphragm	keeps fuel level
Stabilizer	meters fuel

Level 2 version of the gen ed "Carburetor Handout + Carburetor Quiz."
🔑 Key written by our team from the class teacher's materials, please check.

✅ Carburetion & Fuel Quiz · ANSWER KEY

Small Engines Overhaul · Unit 5 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 💧 The main job of a carburetor is… Carb Quiz

A. Making spark | **B. Mixing air and fuel** | C. Cooling oil

2. ⏳ A venturi is… Carb Handout

A. A fuel tank | B. A spark part | **C. A narrow spot that speeds up air**

3. 📉 Fast-moving air has… Carb Handout

A. Lower pressure | B. Higher pressure | C. No pressure

4. 🫧 Which carburetor needs a fuel pump / diaphragm? Carb Quiz

A. Float type | B. Gravity type | **C. Diaphragm type**

5. 🚪 The choke is used for… Carb Handout

A. Cold starting | B. Stopping | C. Cooling

6. 🎚️ The throttle controls… Carb Handout

A. Oil level | **B. Speed** | C. Spark gap

7. 🛟 The float… Carb Handout

A. Makes noise | **B. Shuts off fuel when the bowl is full** | C. Holds oil

8. 🌽 Ethanol in fuel can… Fuel topics

A. Attract water and gum carburetors | B. Make engines last forever | C. Replace oil

9. 🛢️ A 2-stroke needs… Fuel topics

A. No oil | B. Diesel | **C. Oil mixed in the fuel**

10. 🌬️ Blowing across a straw in water lifts the water because… Venturi Lab

A. Fast air lowers pressure | B. Water is light | C. Magic

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed “Carburetor Quiz (paper)”. Score: __ / 10. 🔑 Key written by our team from Mr. Davis's materials, please check.

🎬 Carburetor Videos · ANSWER KEY

Small Engines Overhaul · Unit 5 · Level 2 video guide (modified). Two YouTube videos linked in Canvas (watch with captions).

Name: _____

Before you watch

What do you think a carburetor does? _____

While you watch: circle the best answer

1. What is the first step to remove the carburetor?

A. Start the engine

B. Shut off/drain fuel and remove the air filter

C. Remove the piston

2. What cleans the jets?

A. Water

B. Sand

C. Carburetor cleaner and air

3. What part sets the fuel level?

A. The float

B. The choke

C. The muffler

4. Why take photos while taking it apart?

A. For fun

B. No reason

C. To put the linkages back right

After you watch: finish the sentence

5. After watching, finish the sentence:

A dirty carburetor makes the engine **hard to start or run rough**

Goes with the class "Carburetor Video Handout" (class points pts). Same points when complete.
🔑 Key written by our team from the class teacher's materials, please check.